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February 16, 1967
SUPERSEDING
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FEDERAL SPECIFICATION

PACKING MATERIAL, SEWER JOINT, ASPHALT-SATURATED CELLULOSE-FIBER

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers a rope-like, asphalt-impregnated cellulose-fiber, packing material.

1.2 Classification. The packing material shall be of the following diameter, as specified (see 6.2):

Inch:

3/8

1/2

3/4

7/8

1

1 1/4

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

SS–R–406——Road and Paving Materials; Methods of Sampling and Testing.

PPP–B–601——Boxes, Wood, Cleated-Plywood.

PPP–B–621——Boxes, Wood, Nailed and Lock-Corner.

PPP–B–636——Box, Fiberboard.

PPP–B–640——Boxes, Fiberboard, Corrugated, Triple-Wall.

PPP–T–76——Tape, Pressure-Sensitive Adhesive Paper, (for Carton Sealing).

Federal Standard:

Fed. Std. No. 123——Marking for Domestic Shipment (Civilian Agencies).

FSC 5330

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(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Military Specifications:

MIL–B–121——Barrier Material, Greaseproofed, Waterproofed, Flexible.

Military Standards:

MIL–STD–105——Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129——Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Official Classification Committee (OCC) Publication:

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Ave. at 33rd St., New York, N. Y., 10016.)

American Trucking Association (ATA) Publication:

National Motor Freight Classification Rules.

(Application for copies should be addressed to the American Trucking Association, 1616 P St., N.W., Washington, D. C., 20036.)

3. REQUIREMENTS

3.1 Description. The packing material shall consist of a fibrous carrier saturated with asphalt and shall be flexible, uniform in diameter, and suitable for packing the annular space between the bells and spigots of sewer pipe. The carrier shall consist of matted, fulled, or felted cellulose fibers rolled into shape and securely bound with cords helically wound both clockwise and counterclockwise. The helical winding shall be spaced not more than ¼ inch, measured circumferentially around the packing, and not more than ½ inch measured longitudinally along the packing, and shall pass over and under the windings running in the opposite direction, passing over not more than two successive windings and then under not more than the next two windings. Packing diameters shall be as listed (see 1.2), with a tolerance of plus or minus 1/32 inch.

3.2 Material. Material shall be as specified herein. Materials not specified shall be selected by the supplier and shall be subject to all provisions of this specification.

3.3 Quantitative requirements. The quantitative requirements of the packing shall conform to table I:

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Table I. Quantitative requirements

Property	Requirement	
	Minimum	Maximum
Asphalt (soluble in CS_2) percent of total weight	80	90
Fibrous material (insoluble in CS_2) percent of total weight	10	20
Specific gravity of asphalt at 77°F.	0.90	1.00
Flashpoint of asphalt, C.O.C., °F.	550	—
Weight of packing, grams per linear foot:		
⅜-inch diameter	12	17
½-inch diameter	20	30
¾-inch diameter	50	70
⅞-inch diameter	70	90
1-inch diameter	95	115
1¼-inch diameter	150	180

3.4 Chemical resistance. No effervescence, discoloration of solution, or slimy appearance on the surface of the specimen shall occur when packing is immersed for 5 days in each of the following solutions: 1 percent HCL, 1 percent HNO_3, 1 percent H_2SO_4, 5 percent KOH, and saturated H_2S.

3.5 Flexibility. The packing shall not break or crack when flexed through an arc of 180° around a mandrel 1 inch in diameter.

3.6 Drip and wet spots. The packing shall show no wet spots or shall not drip saturant after 24 hours at 140°F.

3.7 Coils. Packing material shall be supplied in coils containing 50 pounds plus or minus ½ pound of packing material each. Coils shall contain not more than two pieces and no piece shall be less than 10 feet long.

3.8 Workmanship. Packing material shall be uniform in color and texture and shall be free from frayed surfaces which might affect its use for the purpose intended.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection.

(a) Quality conformance inspection (see 4.3).

(b) Inspection of preparation for delivery (see 4.5).

4.3 Quality conformance inspection.

4.3.1 Unit of product. For the purpose of inspection, a unit of product shall be that portion of the first 6-foot segment of each coil in the sample, necessary for the required number of tests and examina-

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tions. For testing packing material as specified in 4.4.2.5, the unit of product shall be the remainder of the 6-foot segment of each coil, necessary for the required number of specimens.

4.3.2 Lot. Each lot shall consist of units of product of a single diameter, manufactured under essentially the same conditions and at essentially the same time.

4.3.3 Sampling. Sampling for examination and tests shall be in accordance with MIL–STD–105, inspection level II.

4.3.4. Examination. Samples selected in accordance with 4.3.3. shall be examined as specified in 4.4.1. AQL shall be 6.5 percent defective for major defects and 15 percent defective for minor defects. A sample having one or more defects shall be considered defective.

4.3.5. Tests. Samples selected in accordance with 4.3.3 shall be tested as specified in 4.4.2. AQL shall be 2.5 percent defective. A sample having failed one or more test shall be considered defective.

4.4 Inspection procedure.

4.4.1 Examination. The packing material shall be examined for the following defects:

Major

101. Binding cords not as specified.

102. Dimensions not as specified.

103. Material not as specified.

Minor

201. Weight and length of packing material per coil not as specified.

4.4.2 Tests.

4.4.2.1 Asphalt. Each specimen shall be tested in accordance with table II. Nonconformance to table I shall constitute failure of the test.

Table II. Test methods

Test	Method in Federal Specification SS–R–406 [1]
Asphalt (soluble in CS_2)	210.0
Fibrous material (insoluble in CS_2)	210.0
Specific gravity of asphalt	209.2
Flashpoint of asphalt	217.01

[1] Applicable ASTM and AASHO methods listed in section 3 of SS–R–406 may be substituted for the methods listed in table II.

4.4.2.2 Weight per linear foot. A specimen 1 foot plus or minus 1/16 inch long of each diameter shall be weighed to the nearest gram. Nonconformance to table I shall constitute failure of this test.

4.4.2.3 Flexibility.

4.4.2.3.1 Preparation of test panel and mandrel. A specimen of packing material 12-inches long and a mandrel 1 inch in diameter shall be conditioned prior to testing in an atmosphere of 40° ± 2°F. and a relative humidity of 90± 5 percent.

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4.4.2.3.2 Procedure. After the conditioning period, the specimen shall be immediately bent through an arc of 180° around the mandrel. Breaking or cracking of the specimen shall constitute failure of this test.

4.4.2.4 Wet spots and drip. A specimen of packing material 6 inches long shall be suspended vertically from a rack. The specimens and rack shall be placed in an oven at 140° ± 2°F. for a period of 24 hours. After this period, the rack and specimens shall be removed from the oven and the specimens shall be examined. Wet spots on or saturant dripping from the specimen shall constitute failure of this test.

4.4.2.5 Chemical resistance. Five specimens, each 2 inches in length, shall be prepared. Each specimen shall be immersed at room temperature for 5 days in one of the five solutions specified in 3.4. Observations shall be made during the 5-day period, and at the end of the 5 days, the specimens and solutions shall be examined. Effervescence or discoloration of solution or slimy appearance on the surface of the specimens shall constitute failure of this test.

4.5 Inspection of preparation for delivery.

4.5.1 Quality conformance inspection of pack.

4.5.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product

4.5.1.2 Sampling. Sampling for examination shall be in accordance with MIL–STD–105. AQL shall be 2.5 percent defective.

4.5.1.3 Examination. Samples selected in accordance with 4.5.1.2 shall be examined for the following defects. A pack having one or more defects shall be considered defective.

104. Materials and containers not as specified for level A or B. Each incorrect material or container shall be considered one defect.
105. Packing material not wrapped as specified for level A or B.
106. Intermediate packaging not as specified for level A or B.
107. Gross weight exceeds 250 pounds for level A.
108. Strapping not zinc coated for level A.
109. Gross weight exceeds the limitation of the box specification for level B.
110. Boxes not reinforced with tape as specified for level B.
111. Marking illegible, incorrect, or incomplete.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A. Each coil of packing material shall be wrapped with barrier material conforming to MIL–B–121, type I, grade A, class 1 or 2, secured by heat sealing or with tape conforming to PPP–T–76.

5.1.1.1 Intermediate packaging. Each wrapped coil of packing material shall be intermediate packaged in a close-fitting box conforming to PPP–B–636, W5c, style optional.

5.1.2 Level B. Unless otherwise specified (see 6.2 and 6.3), each coil of packing material shall be wrapped and intermediate packaged as specified in 5.1.1 and 5.1.1.1 for level A except the box for intermediate packaging shall be type CF, class domestic, variety SW, grade 200.

5.1.3 Level C. Each coil of packing material shall be packaged in a manner to afford protection against deterioration and damage from the supplier to the initial destination. The supplier may use his standard practice when it meets these requirements.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. The packing material, packaged as specified in 5.1, shall be packed in close-fitting boxes conforming to PPP–B–601, overseas type, style optional, or PPP–B–621, class 2, style optional.

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The gross weight of each box shall not exceed 250 pounds. The boxes shall be closed and strapped in accordance with the appendix to the applicable box specification. Strapping shall be zinc coated.

5.2.2 Level B. The packing material, packaged as specified in 5.1, shall be packed in close-fitting boxes conforming to PPP–B–636, V3c, style optional, or PPP–B–640, class 2, grade A, style optional. The gross weight or size shall not exceed the weight limitation of the box specification. The boxes shall be reinforced with bands of reinforced filament tape applied in accordance with the appendix to the box specification.

5.2.3 Level C. The packing material, packaged as specified in 5.1, shall be packed to assure carrier acceptance and safe delivery to destination at lowest rates in fiberboard containers complying with Uniform Freight Classification Rules, National Motor Freight Classification Rules, or with other carrier rules applicable to the mode of transportation.

5.3 Marking.

5.3.1 Civil agencies. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL–STD–129.

6. NOTES

6.1 Intended use. The packing material covered by this specification is intended for packing a portion of the annular space between the bells and spigots of sewer pipe prior to filling with bituminous joint sealing compound of the cold-applied mastic type.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(*a*) Title, number, and date of this specification.

(*b*) Diameter of packing material required (see 1.2).

(*c*) Selection of applicable levels of packaging and packing required (see 5.1 and 5.2).

6.3 Level B packaging (see 5.1.2), has been developed for use by civil agencies, Army, and Air Force for the specific conditions specified in 6.2. For Navy procurements, details of level B packaging when required, should be developed at the time of procurement when all conditions are known.

CUSTODIANS:

Army—MO(ERDL)
Navy—YD
Air Force—69

Review activities:

Army—MO, CE
Navy—YD
Air Force—69

User activities:

Army—GL
Navy—CG
Air Force—None

Civil activity:

DC—GSA

Preparing activity:

Army—MO(ERDL)

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Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price cents each.

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